A COMPLEX OF RIGHT ACYCLIC QUASI-COHERENT MODULES REPRESENTS ITS OWN DERIVED PUSHFORWARD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The pushforward of a bounded below complex of right acyclic quasi-coherent modules computes the derived pushforward. invoked at proof lines 751-754. Finishes the proof once the complex has been made bounded below.
- The zeroth higher direct image of the complex agrees with that of a suitable truncation from below, because of quasi-coherence of the direct image of the terms. invoked at proof lines 745-749. Lets the proof replace the complex by a bounded below one without changing the degree zero computation.

Source reference. Original lemma 08D5: sources/stacks-perfect/pinned-original-source.tex, source lines 661–661. Invoked at proof lines 745-748 to identify the zeroth higher direct image of the complex with that of a truncation from below; the reduction that uses this is inside the counted span.

Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 751-754 to conclude that the comparison map is a quasi-isomorphism; the reductions that make it applicable are inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a quasi-separated and quasi-compact morphism of schemes. Let $\mathcal{F}^\bullet$ be a complex of quasi-coherent $\mathcal{O}_X$-modules each of which is right acyclic for f_*. Then $f_*\mathcal{F}^\bullet$ represents $Rf_*\mathcal{F}^\bullet$ in $D(\mathcal{O}_S)$.*

Proof. There is always a canonical map $f_*\mathcal{F}^\bullet \to Rf_*\mathcal{F}^\bullet$. Our task is to show that this is an isomorphism on cohomology sheaves. As the statement is invariant under shifts it suffices to show that $H^0(f_*(\mathcal{F}^\bullet)) \to R^0f_*\mathcal{F}^\bullet$ is an isomorphism. The statement is local on S hence we may assume S affine. By Lemma 08D5 we have $R^0f_*\mathcal{F}^\bullet = R^0f_*\tau_{\geq -n}\mathcal{F}^\bullet$ for all sufficiently large n. Thus we may assume $\mathcal{F}^\bullet$ bounded below. As each $\mathcal{F}^n$ is right f_*-acyclic by assumption we see that $f_*\mathcal{F}^\bullet \to Rf_*\mathcal{F}^\bullet$ is a quasi-isomorphism by Leray's acyclicity lemma (Derived Categories, Lemma 015E). □

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